STAD29 / STA 1007 assignment 7

Due Tuesday Mar 26 at 11:59pm on Quercus

Packages for this one:

```
library(tidyverse)

## -- Attaching packages --------------------------------- tidyverse 1.2.1 --
## v ggplot2 3.1.0     v purrr   0.3.1
## v tibble  2.0.1     v dplyr   0.8.0.1
## v tidyr   0.8.3     v stringr 1.4.0
## v readr   1.3.1     v forcats 0.3.0
## -- Conflicts ------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

Hand in problems 2 and 4.

1. Work through Problems 23.1-23.4 and 23.8 of PASIAS.

2. Some data were collected on 20 different animals. For each animal, it was recorded whether or not the animal:

   - is warm-blooded
   - can fly
   - is a vertebrate (has a backbone)
   - is endangered
   - lives in groups
   - has hair

   For each pair of animals, the dissimilarity was calculated as the number of properties above on which the animals *disagree*. The data are in `https://www.utsc.utoronto.ca/~butler/d29/animal-dissim.csv`.

   (a) (2 marks) Read in the data and display at least some of it.

   (b) (2 marks) Turn your data frame into a `dist` object and display it.

   (c) (3 marks) Run a hierarchical cluster analysis using complete linkage, and display a dendrogram.

   (d) (4 marks) Pick two clusters, containing at least two animals in each, that get joined together somewhere in your dendrogram. Verify that the "height" at which the two clusters are joined together in your dendrogram is correct.

   (e) (2 marks) Obtain a cluster analysis using Ward's method, and display the results on a dendrogram.

   (f) (3 marks) What do you think is a good number of clusters? Add these clusters to your plot.

   (g) (2 marks) Based on what you know about these animals, what do the ones in the same one of your clusters have in common (or, equivalently, what makes your clusters different)?

3. Work through Problems 23.5-23.7 and 23.9 of PASIAS.

4. We previously looked at a dataset on wine quality. The data are in `http://www.utsc.utoronto.ca/~butler/d29/wine_quality.csv`. What we did before was to see which of the quantitative variables were related to the wine quality. This time, we are going to see how the wines are like *each other* using a cluster analysis, without looking at the quality ratings at all (until near the end).

   (a) (2 marks) Read in and display some of the data.

   (b) (4 marks) Get rid of the column containing the quality ratings and standardize everything else (with `scale` or otherwise). Save the result. Display some of it. Given that some of the variables are skewed to the right, does it look as if you successfully standardized your columns, based on what you see? Explain briefly.

   (c) (3 marks) Obtain a K-means clustering using 4 clusters, and save the result. Don't display it yet. Use the option that allows K-means to run multiple times and take the best answer.

   (d) (2 marks) Make a data frame containing the wine quality ratings from the original data set and the cluster memberships from the `kmeans` output. Display the first few lines of this data frame.

   (e) (4 marks) Make a table showing how many wines from each quality rating (columns) ended up in each cluster (rows). Which cluster did most of the Excellent wines end up in? Do you think that the clusters and the quality ratings correspond well? Explain briefly.

   (f) (4 marks) Make a scree plot for these data, for each number of clusters from 2 to 20. Did it make sense to fit four clusters, or would a different number have been better? (The data set is big, so the calculation might be slow.) You might get a warning about "did not converge in 10 iterations"; if that happens to you, add something like `iter.max=20` to your `kmeans`.

Notes

[1] This is because rows and columns are different kinds of things in data analysis.

[2] I got this right first time, but then I knew exactly what I was doing. When you do it, get each line right before moving on to the next.